

1. Dataset Shape

Marketing_Campaign_Analysis.ipynb

Colab output

```
[7] import pandas as pd

df = pd.read_csv("marketing_campaign_data.csv")

print(df.shape)

df.head()
```

(56000, 28)

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	\
0	342199	1985	Graduation	Together	59011.7	1	
1	8075450	1975	Master	Single	1730.0	1	
2	13664263	1978	Graduation	Married	98584.6	0	
3	16164787	1976	Graduation	Married	74031.5	1	
4	15815139	1981	Graduation	Divorced	52784.2	1	

	Teenhome	Dt_Customer	Recency	MntWines	...	NumStorePurchases	\
0	0	2012-11-17	3	0	...	3	
1	1	2013-04-10	96	0	...	2	
2	0	2014-01-11	99	920	...	6	
3	0	2014-06-18	47	265	...	11	
4	1	2014-05-20	0	30	...	3	

	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	\
0	4	0	0	0	0	
1	3	0	0	0	0	
2	3	0	0	0	0	
3	4	0	0	0	0	
4	6	0	0	0	0	

	AcceptedCmp2	Response	Complain	Country
0	0	0	0	Spain
1	0	0	0	Spain
2	0	0	0	Australia
3	0	0	0	Spain
4	0	0	0	Canada

[5 rows x 28 columns]

2. Dataset Info

Marketing_Campaign_Analysis.ipynb

Colab output

[9] df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 56000 entries, 0 to 55999
Data columns (total 28 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    56000 non-null  int64
1   Year_Birth            56000 non-null  int64
2   Education              56000 non-null  object
3   Marital_Status        56000 non-null  object
4   Income                56000 non-null  float64
5   Kidhome               56000 non-null  int64
6   Teenhome              56000 non-null  int64
7   Dt_Customer           56000 non-null  object
8   Recency               56000 non-null  int64
9   MntWines              56000 non-null  int64
10  MntFruits              56000 non-null  int64
11  MntMeatProducts        56000 non-null  int64
12  MntFishProducts        56000 non-null  int64
13  MntSweetProducts       56000 non-null  int64
14  MntGoldProds           56000 non-null  int64
15  NumDealsPurchases      56000 non-null  int64
16  NumWebPurchases        56000 non-null  int64
17  NumCatalogPurchases    56000 non-null  int64
18  NumStorePurchases      56000 non-null  int64
19  NumWebVisitsMonth      56000 non-null  int64
20  AcceptedCmp3           56000 non-null  int64
21  AcceptedCmp4           56000 non-null  int64
22  AcceptedCmp5           56000 non-null  int64
23  AcceptedCmp1           56000 non-null  int64
24  AcceptedCmp2           56000 non-null  int64
25  Response               56000 non-null  int64
26  Complain               56000 non-null  int64
27  Country                56000 non-null  object
dtypes: float64(1), int64(23), object(4)
memory usage: 12.0+ MB
```

3. Missing Values Handling

Marketing_Campaign_Analysis.ipynb

Colab output

```
[10] # Missing values

df.isnull().sum()

# Fill missing values

for col in df.columns:

    if df[col].dtype == "object":
        df[col] = df[col].fillna(df[col].mode()[0])

    else:
        df[col] = df[col].fillna(df[col].median())

print("Missing values handled")

Missing values handled
```

4. Feature Engineering

Marketing_Campaign_Analysis.ipynb

Colab output

```
[11] # Age

df["Age"] = 2026 - df["Year_Birth"]

# Children

df["Children"] = df["Kidhome"] + df["Teenhome"]

# Total Spend

df["Total_Spend"] = (
    df["MntWines"] +
    df["MntFruits"] +
    df["MntMeatProducts"] +
    df["MntFishProducts"] +
    df["MntSweetProducts"] +
    df["MntGoldProds"]
)

# Total Purchases

df["Total_Purchases"] = (
    df["NumWebPurchases"] +
    df["NumCatalogPurchases"] +
    df["NumStorePurchases"]
)

print("Features Created Successfully")
```

Features Created Successfully

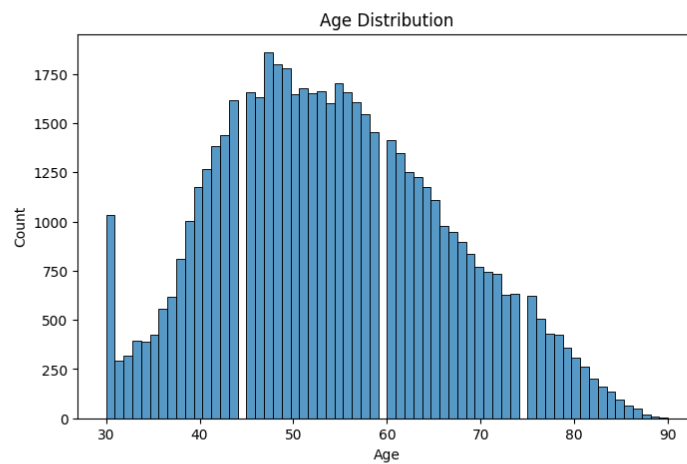
5. Age & Income Distribution

Marketing_Campaign_Analysis.ipynb

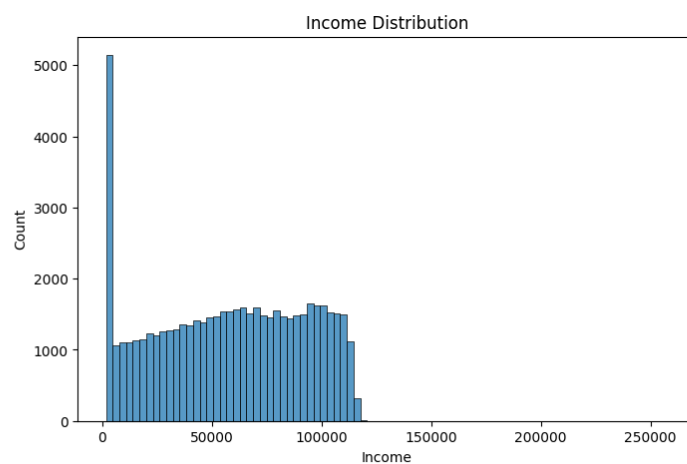
Colab output

```
[12] import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))
sns.histplot(df["Age"])
plt.title("Age Distribution")
plt.show()
```



```
[13] plt.figure(figsize=(8,5))
sns.histplot(df["Income"])
plt.title("Income Distribution")
plt.show()
```

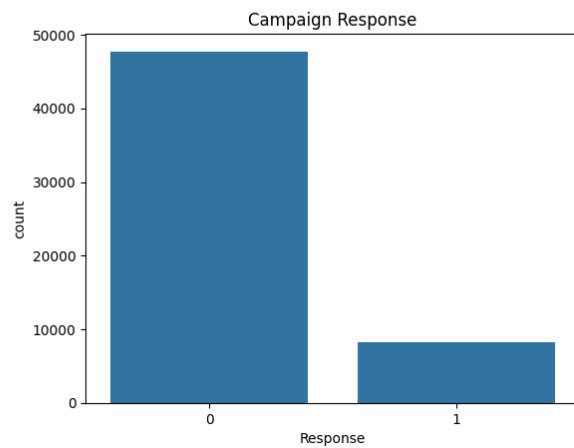


6. Campaign Response

Marketing_Campaign_Analysis.ipynb

Colab output

```
[14] sns.countplot(x="Response", data=df)
plt.title("Campaign Response")
plt.show()
```



7. Customer Segmentation & Saved Dataset

Marketing_Campaign_Analysis.ipynb

Colab output

```
[15] df["Income_Segment"] = pd.cut(
    df["Income"],
    bins=[0,30000,60000,1000000],
    labels=["Low","Medium","High"]
)

threshold = df["Total_Spend"].quantile(0.90)

df["High_Spender"] = (
    df["Total_Spend"] > threshold
).astype(int)

df["High_Web_Engagement"] = (
    df["NumWebVisitsMonth"] > 5
).astype(int)

df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	\
0	342199	1985	Graduation	Together	59011.7	1	
1	8075450	1975	Master	Single	1730.0	1	
2	13664263	1978	Graduation	Married	98584.6	0	
3	16164787	1976	Graduation	Married	74031.5	1	
4	15815139	1981	Graduation	Divorced	52784.2	1	

	Teenhome	Dt_Customer	Recency	MntWines	...	Response	Complain	\
0	0	2012-11-17	3	0	...	0	0	
1	1	2013-04-10	96	0	...	0	0	
2	0	2014-01-11	99	920	...	0	0	
3	0	2014-06-18	47	265	...	0	0	
4	1	2014-05-20	0	30	...	0	0	

	Country	Age	Children	Total_Spend	Total_Purchases	Income_Segment	\
0	Spain	41	1	69	10	Medium	
1	Spain	51	2	39	3	Low	
2	Australia	48	0	1512	9	High	
3	Spain	50	1	478	15	High	
4	Canada	45	2	330	8	Medium	

	High_Spender	High_Web_Engagement
0	0	0
1	0	0
2	0	0
3	0	0
4	0	1

[5 rows x 35 columns]

```
[16] df.to_csv(
    "marketing_cleaned.csv",
    index=False
)

print("Dataset Saved")
```

Dataset Saved